

# Chapter 3: Method of Frobenius

## Mathematical Physics

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September 20, 2026

# Outline

- 1 The guitar string and the drum head
- 2 When will the method of Frobenius work?
- 3 Second solutions
- 4 Airy functions in the complex plane

# Where we are going

Separation of variables turns many partial differential equations (PDEs) into linear, second-order, homogeneous ordinary differential equations. Two recurring examples are

- the **guitar string**, which gives the harmonic-oscillator equation

$$\frac{d^2 y}{dx^2} + k^2 y = 0,$$

- the **drum**, which gives Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (k^2 x^2 - m^2) y = 0.$$

Here  $k$  is a wave number, and  $m$  is the nonnegative integer angular order of a drum mode. On an interval where the coefficient of  $y''$  is nonzero, a linear homogeneous second-order equation can be written as

$$y'' + P(x)y' + Q(x)y = 0.$$

Here *linear* means that  $y, y', y''$  occur only to the first power and are not multiplied together; *homogeneous* means that the right-hand side is zero. The method of Frobenius is a local series method for this class of equations. It succeeds at ordinary points and regular singular points; later we state the precise conditions.

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# Guitar string: the harmonic oscillator by Taylor series

**Step 1: insert a power series.** If the physical string of length  $L$  obeys  $\frac{d^2 y}{ds^2} + k^2 y = 0$  on  $0 \leq s \leq L$ , substitute  $x = ks$ ; by the chain rule  $\frac{d^2 y}{ds^2} = k^2 \frac{d^2 y}{dx^2}$ , so the equation becomes  $y'' + y = 0$  and the physical endpoints  $s = 0, L$  map to  $x = 0$  and  $x = kL$ . The fundamental mode has wave number  $k = \pi/L$ ; we justify this choice from the endpoint conditions below. It makes the rescaled interval  $0 \leq x \leq \pi$ , so we first solve

$$y'' + y = 0, \quad y(0) = 0, \quad y(\pi) = 0. \quad (3.1.1)$$

Assume that  $y$  is analytic at  $x = 0$  and write

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots.$$

Term-by-term differentiation gives

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \cdots,$$
$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = 2a_2 + 3 \cdot 2a_3 x + 4 \cdot 3a_4 x^2 + \cdots.$$

# Guitar string: the harmonic oscillator by Taylor series (cont.)

Substitution into (3.1.1) yields

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0.$$

**Step 2: align the powers.** The powers do not yet match. Reindexing aligns every sum on the same power of  $x$ . In the first sum set

$$j = n - 2, \quad n = j + 2.$$

When  $n = 2$ ,  $j = 0$ , and as  $n \rightarrow \infty$ ,  $j \rightarrow \infty$ . Therefore

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} &= \sum_{j=0}^{\infty} (j+2)(j+1)a_{j+2} x^j \\ &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n, \end{aligned}$$

# Guitar string: the harmonic oscillator by Taylor series (cont.)

where the last line only renames the dummy index  $j$  as  $n$ . Hence

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n] x^n = 0.$$

**Step 3: read off the recurrence.** A power series is identically zero only when each coefficient is zero. Thus, for every  $n \geq 0$ ,

$$(n+2)(n+1)a_{n+2} + a_n = 0,$$

and solving for the next coefficient gives

$$a_{n+2} = -\frac{a_n}{(n+1)(n+2)}, \quad n = 0, 1, 2, \dots \quad (3.1.2)$$

The recurrence advances the index by two. Consequently,  $a_0$  generates all even coefficients, while  $a_1$  generates all odd coefficients.

# Resolving the even recurrence into $\cos x$

Starting with  $a_0$ , the recurrence (3.1.2) gives

$$a_2 = -\frac{a_0}{1 \cdot 2} = -\frac{a_0}{2!}, \quad a_4 = -\frac{a_2}{3 \cdot 4} = \frac{a_0}{4!}.$$

Each step introduces the next two factors and reverses the sign. Hence

$$a_{2j} = \frac{(-1)^j a_0}{(2j)!}, \quad j = 0, 1, 2, \dots$$

The even part of the solution is therefore

$$a_0 \left( 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) = a_0 \cos x.$$

This is the cosine Taylor series from Chapter 1, now recovered from the differential equation.



# Resolving the odd recurrence into $\sin x$

Starting with  $a_1$  instead,

$$a_3 = -\frac{a_1}{2 \cdot 3} = -\frac{a_1}{3!}, \quad a_5 = -\frac{a_3}{4 \cdot 5} = \frac{a_1}{5!}.$$

The pattern is

$$a_{2j+1} = \frac{(-1)^j a_1}{(2j+1)!}, \quad j = 0, 1, 2, \dots,$$

so the odd part is  $a_1(x - x^3/3! + x^5/5! - \dots) = a_1 \sin x$ . Both starting coefficients are free, giving

$$y(x) = a_0 \cos x + a_1 \sin x, \quad a_0 = y(0), \quad a_1 = y'(0).$$

The differential equation determines the shape; endpoint or initial conditions determine which combination is needed.

# Applying the endpoint conditions

The string is fixed at its left end, so  $y(0) = a_0 = 0$ . In the scaled coordinate  $x = ks$ , the right end  $s = L$  is  $x = kL$ . Thus

$$0 = y(kL) = a_1 \sin(kL).$$

A nonzero mode has  $a_1 \neq 0$ , so

$$kL = n\pi, \quad n = 1, 2, \dots$$

The fundamental mode is  $n = 1$ , which justifies the choice  $k = \pi/L$  at the start. In its scaled interval  $0 \leq x \leq \pi$ ,

$$y(x) = A \sin x.$$

The endpoint conditions do not determine the amplitude  $A$ ; an additional amplitude or normalization condition is needed.

# Drum: Bessel's equation needs a generalized series

**Step 1: scale the radial equation.** For a circular drum, separation into angular modes gives the following radial equation, which we take as our model:

$$\rho^2 Y''(\rho) + \rho Y'(\rho) + (k^2 \rho^2 - m^2) Y(\rho) = 0.$$

Here  $\rho$  is physical radius,  $k > 0$  is a wave number, and the nonnegative integer  $m$  labels angular variation. Put  $x = k\rho$  and  $Y(\rho) = y(x)$ . The chain rule gives

$$Y'(\rho) = ky'(x), \quad Y''(\rho) = k^2 y''(x).$$

Substitution, using  $k\rho = x$ , yields Bessel's normalized equation

$$x^2 y'' + xy' + (x^2 - m^2)y = 0. \quad (3.1.3)$$

The physical rim  $\rho = R$  is now  $x = kR$ ; this is where  $k$  returns in the boundary condition  $J_m(kR) = 0$ .

For the drum, the angular order is  $m = 0, 1, 2, \dots$ . Because  $x = 0$  is a singular point of the normalized equation, a Taylor series need not capture every solution. On  $x > 0$ , use the Frobenius ansatz

$$y(x) = x^r \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0. \quad (3.1.4)$$

## Drum: Bessel's equation needs a generalized series (cont.)

The first two derivatives are

$$y'(x) = \sum_{k=0}^{\infty} (k+r) a_k x^{k+r-1},$$
$$y''(x) = \sum_{k=0}^{\infty} (k+r)(k+r-1) a_k x^{k+r-2}.$$

No lower-limit terms should be deleted merely because  $r$  is unknown: a coefficient such as  $r(r-1)$  may or may not vanish after the indicial equation is solved.

**Step 2: substitute and align powers.** Substitution into (3.1.3) gives

$$0 = x^2 \sum_{k=0}^{\infty} (k+r)(k+r-1) a_k x^{k+r-2} + x \sum_{k=0}^{\infty} (k+r) a_k x^{k+r-1} \\ + (x^2 - m^2) \sum_{k=0}^{\infty} a_k x^{k+r}.$$

# Drum: Bessel's equation needs a generalized series (cont.)

Distributing the powers of  $x$  yields

$$\begin{aligned} 0 = & \sum_{k=0}^{\infty} (k+r)(k+r-1)a_k x^{k+r} + \sum_{k=0}^{\infty} (k+r)a_k x^{k+r} \\ & + \sum_{k=0}^{\infty} a_k x^{k+r+2} - m^2 \sum_{k=0}^{\infty} a_k x^{k+r}. \end{aligned}$$

Only the third sum has the shifted power  $x^{k+r+2}$ . Set

$$j = k + 2, \quad k = j - 2.$$

Then

$$\begin{aligned} \sum_{k=0}^{\infty} a_k x^{k+r+2} &= \sum_{j=2}^{\infty} a_{j-2} x^{j+r} \\ &= \sum_{k=2}^{\infty} a_{k-2} x^{k+r}. \end{aligned}$$

## Drum: Bessel's equation needs a generalized series (cont.)

*Reindexing aligns every sum on the same power of  $x$ .* Introduce the bookkeeping convention

$$a_{-2} = a_{-1} = 0.$$

It allows the shifted sum to begin at  $k = 0$  without adding nonzero terms:

$$\sum_{k=2}^{\infty} a_{k-2} x^{k+r} = \sum_{k=0}^{\infty} a_{k-2} x^{k+r}.$$

**Step 3: read off the indicial equation.** Therefore

$$\sum_{k=0}^{\infty} [(k+r)(k+r-1)a_k + (k+r)a_k - m^2 a_k + a_{k-2}] x^{k+r} = 0.$$

Since

$$(k+r)(k+r-1) + (k+r) = (k+r)[(k+r-1) + 1] = (k+r)^2,$$

the coefficient equation is

$$[(k+r)^2 - m^2] a_k + a_{k-2} = 0, \quad k = 0, 1, 2, \dots \quad (3.1.5)$$

## Drum: Bessel's equation needs a generalized series (cont.)

At  $k = 0$ ,  $a_{-2} = 0$ , so

$$(r^2 - m^2)a_0 = 0.$$

Because  $a_0 \neq 0$ , the indicial equation is

$$r^2 - m^2 = 0 \iff (r - m)(r + m) = 0, \quad (3.1.6)$$

and hence

$$r = m \quad \text{or} \quad r = -m.$$

The root  $r = m \geq 0$  gives the solution that is finite at the origin.

# The Bessel recurrence and $J_m(x)$

Choose the regular root  $r = m$ . Since  $(k + m)^2 - m^2 = k(k + 2m)$ , (3.1.5) becomes

$$k(k + 2m)a_k + a_{k-2} = 0. \quad (3.1.7)$$

At  $k = 1$ ,  $(1 + 2m)a_1 = 0$ , so  $a_1 = 0$ . Every later odd coefficient is zero. For the even coefficients,

$$a_2 = -\frac{a_0}{2(2 + 2m)} = -\frac{a_0}{4(m + 1)}, \quad a_4 = -\frac{a_2}{4(4 + 2m)} = \frac{a_0}{32(m + 1)(m + 2)}.$$

At the  $j$ th step the denominator factor is  $(2j)(2j + 2m) = 4j(j + m)$ , so iterating  $a_{2j} = -a_{2j-2}/[4j(j + m)]$  from  $a_0$  gives

$$a_{2j} = (-1)^j a_0 \prod_{i=1}^j \frac{1}{4i(i + m)} = \frac{(-1)^j a_0}{4^j j! \prod_{i=1}^j (i + m)}.$$

Since  $\prod_{i=1}^j (i + m) = (m + 1)(m + 2) \cdots (m + j) = (m + j)!/m!$  and  $4^j = 2^{2j}$ ,

$$a_{2j} = \frac{(-1)^j m! a_0}{2^{2j} j! (m + j)!}, \quad a_{2j+1} = 0. \quad (3.1.8)$$



# The Bessel recurrence and $J_m(x)$ (cont.)

The overall scale is still arbitrary. The conventional choice

$$a_0 = \frac{1}{2^m m!}$$

turns the series into the Bessel function of the first kind:

$$J_m(x) = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!(m+j)!} \left(\frac{x}{2}\right)^{2j+m}, \quad m = 0, 1, 2, \dots \quad (3.1.9)$$

Indeed  $a_{2j}x^{2j+m} = (-1)^j(x/2)^{2j+m}/[j!(m+j)!]$  with this choice. Any multiple  $AJ_m(x)$  is also a regular solution. Its amplitude will be fixed by the physical data, not by the name  $J_m$ .

# What the other Bessel root is trying to say

The root  $r = +m$  gave the solution that is finite at the center of the drum. What happens to the other root?

For a positive noninteger order  $\nu$ , the same coefficient calculation, with factorials replaced by Gamma functions, gives

$$J_\nu(x) = \sum_{j=0}^{\infty} \frac{(-1)^j}{j! \Gamma(j + \nu + 1)} \left(\frac{x}{2}\right)^{2j+\nu}, \quad x > 0.$$

Replacing  $\nu$  by  $-\nu$  gives  $J_{-\nu}$ . Their first terms are

$$J_\nu(x) \sim \frac{x^\nu}{2^\nu \Gamma(\nu + 1)}, \quad J_{-\nu}(x) \sim \frac{2^\nu x^{-\nu}}{\Gamma(1 - \nu)}.$$

One tends to zero and the other diverges as  $x \rightarrow 0^+$ , so they cannot be constant multiples of each other. They are the two independent solutions.

# What the other Bessel root is trying to say (cont.)

**Integer orders need a different second solution.** For the negative root  $r = -\nu$ , the coefficient equation is

$$k(k - 2\nu)a_k + a_{k-2} = 0. \quad (3.1.10)$$

Before dividing, check whether  $k(k - 2\nu)$  can vanish. For example, at order  $m = 1$ ,

$$k(k - 2)a_k + a_{k-2} = 0.$$

At  $k = 2$  this requires  $0 \cdot a_2 + a_0 = 0$ , contradicting  $a_0 \neq 0$ . The assumed pure series cannot supply the missing solution. The same obstruction occurs for every positive integer  $m$ ; for  $m = 0$  the two roots coincide. The second solution is called  $Y_m$  and is singular at the origin. We construct it by reduction of order later in this chapter.

# What the other Bessel root is trying to say (cont.)

**The half-order example reduces to familiar functions.** For  $\nu = 1/2$  and  $r = -1/2$ , the coefficient equation becomes

$$k(k-1)a_k + a_{k-2} = 0.$$

The equations at  $k = 0, 1$  leave  $a_0, a_1$  free. For  $k \geq 2$ ,

$$a_k = -\frac{a_{k-2}}{k(k-1)},$$

which is exactly the even-odd recurrence of the harmonic oscillator. Therefore

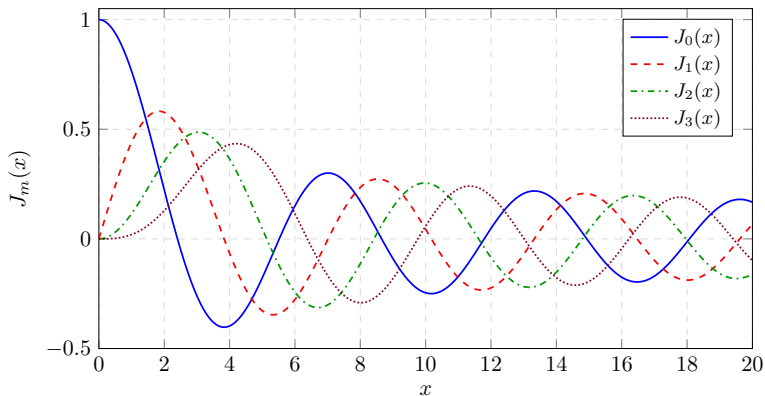
$$y(x) = x^{-1/2}(a_0 \cos x + a_1 \sin x).$$

Match each branch to a root by its leading power:  $x^{-1/2} \cos x$  has leading power  $x^{-1/2}$  (the root  $r = -1/2$ ), while  $x^{-1/2} \sin x \sim x^{1/2}$  (the root  $r = +1/2$ ). Comparing with the normalized leading term  $J_\nu \sim (x/2)^\nu / \Gamma(\nu + 1)$  and using  $\Gamma(1/2) = \sqrt{\pi}$ ,  $\Gamma(3/2) = \sqrt{\pi}/2$  fixes both amplitudes to  $\sqrt{2/\pi}$ :

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x, \quad J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x.$$

Thus a special function may be an elementary function at particular orders.

# The Bessel functions of the first kind $J_0, \dots, J_3$



Near the origin,  $J_0(0) = 1$ , while  $J_m(0) = 0$  for every integer  $m \geq 1$ . The next frame states the corresponding local and large- $x$  formulas explicitly.

# Bessel asymptotics and drum eigenvalues

The first two terms of the defining series (3.1.9) give, for fixed  $m$ ,

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m \left[1 - \frac{x^2}{4(m+1)} + O(x^4)\right] \quad (x \rightarrow 0).$$

Thus  $J_0 \rightarrow 1$ , while  $J_m$  vanishes like  $x^m$  for  $m \geq 1$ . For fixed  $m$  and large positive  $x$ ,

$$J_m(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}),$$

(derived by stationary phase in Chapter 5), so the oscillation amplitude decays like  $x^{-1/2}$ . If a drum has radius  $R$ , the rim condition is  $J_m(kR) = 0$ . Writing  $\alpha_{m,n}$  for the  $n$ th positive zero of  $J_m$  (the notation used again in Chapter 5), the allowed radial wave numbers are

$$k_{m,n} = \frac{\alpha_{m,n}}{R}.$$

# Choose the series by classifying the point

First put the equation in the form  $y'' + P(x)y' + Q(x)y = 0$  and set  $z = x - x_0$ . Here *analytic at  $x_0$*  means representable by a convergent Taylor series there.

- **Ordinary point:**  $P$  and  $Q$  are analytic at  $x_0$ . An ordinary Taylor series supplies both independent solutions.
- **Regular singular point:** the point is not ordinary, but  $zP$  and  $z^2Q$  are analytic at  $z = 0$ . Use a power  $z^r$  times a Taylor series.
- **Irregular singular point:** neither test holds. Frobenius is not guaranteed to work.

We first use this test as a recipe, then examine why it works in the next section.

# The method of Frobenius: the recipe

Let  $z = x - x_0$ . To analyze

$$p_0(x)y'' + p_1(x)y' + p_2(x)y = 0$$

near  $x = x_0$ :

- 1 First divide by  $p_0(x)$  on a punctured neighborhood where  $p_0(x) \neq 0$ , giving  $y'' + P(x)y' + Q(x)y = 0$ .
- 2 Classify  $x_0$ . At an ordinary point, use a Taylor series with free initial coefficients. At a regular singular point, use

$$y(x) = \sum_{k=0}^{\infty} a_k z^{k+r}, \quad a_0 \neq 0.$$

- 3 Differentiate and reindex the sums so that their powers match. Equate coefficients of the same power, as in the string and Bessel examples.
- 4 For a Frobenius series, set the coefficient of the lowest power to zero. This gives the *indicial equation* for  $r$ .



# The method of Frobenius: the recipe (cont.)

- 5 For each admissible root  $r$ , set every remaining coefficient equal to zero and solve the resulting recurrence, checking any index at which its denominator vanishes as a separate compatibility equation.
- 6 Substitute the coefficients back into the series and verify the result in the original differential equation and against any initial or boundary conditions.

A vanishing recurrence denominator must never be crossed by formal division: it can force a coefficient, free a new coefficient, or make a branch inconsistent.

## Example: Airy's equation by series

Airy's equation is

$$y'' - xy = 0. \quad (3.1.11)$$

The sign of  $x$  suggests oscillation for  $x < 0$  and growth or decay for  $x > 0$ . These are qualitative guides; the coefficient is not constant.

At  $x = 0$ , the coefficients  $P = 0$  and  $Q = -x$  are analytic. An ordinary Taylor series is therefore enough. Write

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Differentiate once more, then shift the index so that the power is  $x^n$ :

$$y'' = \sum_{j=2}^{\infty} j(j-1) a_j x^{j-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

For the other term, multiplication by  $x$  shifts the coefficients in the opposite way:

$$xy = \sum_{j=0}^{\infty} a_j x^{j+1} = \sum_{n=1}^{\infty} a_{n-1} x^n.$$

## Example: Airy's equation by series (cont.)

The constant coefficient in  $y'' - xy = 0$  is  $2a_2$ , so  $a_2 = 0$ . For  $n \geq 1$ , equating the coefficients of  $x^n$  gives

$$(n+2)(n+1)a_{n+2} = a_{n-1}. \quad (3.1.12)$$

Set  $k = n - 1$  to write this as a forward recurrence:

$$a_{k+3} = \frac{a_k}{(k+3)(k+2)}, \quad k = 0, 1, 2, \dots \quad (3.1.13)$$

The coefficients separate into three chains. The chain starting at  $a_2$  vanishes. The other two begin

$$a_3 = \frac{a_0}{3 \cdot 2} = \frac{a_0}{6}, \quad a_4 = \frac{a_1}{4 \cdot 3} = \frac{a_1}{12}.$$

Neither  $a_0$  nor  $a_1$  is constrained by the equation.

## Example: Airy's equation by series (cont.)

Define the two series obtained by setting the free coefficients to  $(1, 0)$  and  $(0, 1)$ :

$$f_0(x) = 1 + \frac{x^3}{6} + \frac{x^6}{180} + \cdots, \quad f_1(x) = x + \frac{x^4}{12} + \frac{x^7}{504} + \cdots.$$

Then every solution is

$$y(x) = a_0 f_0(x) + a_1 f_1(x). \quad (3.1.14)$$

The constants are exactly the initial data:

$$a_0 = y(0), \quad a_1 = y'(0),$$

since  $f_0(0) = 1$ ,  $f_0'(0) = 0$ ,  $f_1(0) = 0$ , and  $f_1'(0) = 1$ . Thus the two series can match any prescribed value and slope at zero. The classical Airy functions below are particular combinations of this same pair.

# Standard Airy solutions $Ai$ and $Bi$

Using the basis  $f_0, f_1$ , the classical Airy functions are fixed by these standard initial values:

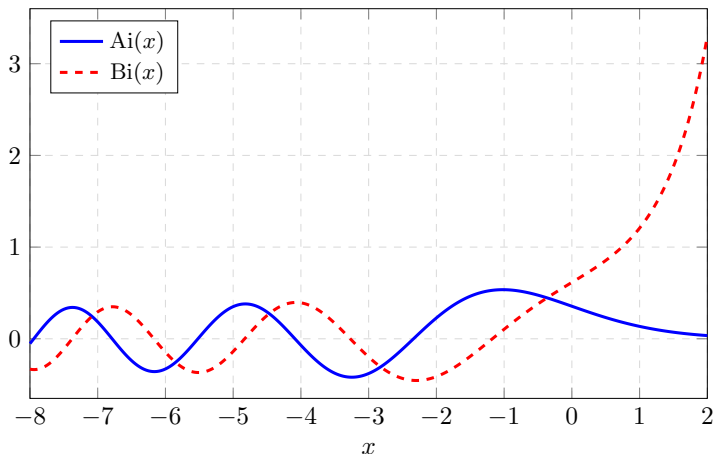
$$\begin{aligned} Ai(0) &= \frac{1}{3^{2/3}\Gamma(2/3)}, & Ai'(0) &= -\frac{1}{3^{1/3}\Gamma(1/3)}, \\ Bi(0) &= \frac{1}{3^{1/6}\Gamma(2/3)} = \sqrt{3} Ai(0), & Bi'(0) &= \frac{3^{1/6}}{\Gamma(1/3)} = -\sqrt{3} Ai'(0). \end{aligned} \tag{3.1.15}$$

(We quote these constants; they come from classical integral representations of  $Ai$  and  $Bi$ , not from the power-series calculation itself.) Explicitly,

$$\begin{aligned} Ai(x) &= Ai(0) f_0(x) + Ai'(0) f_1(x), \\ Bi(x) &= Bi(0) f_0(x) + Bi'(0) f_1(x). \end{aligned} \tag{3.1.16}$$

The two initial-value pairs are not proportional, so every Airy solution can be written  $c_1 Ai + c_2 Bi$ . For large positive  $x$ ,  $Ai$  decays and  $Bi$  grows; for negative  $x$ , both oscillate. These behaviors guide the choice imposed by boundary conditions.

# The Airy functions $Ai$ and $Bi$



The plot shows the classical combinations (3.1.16) of the two power series  $f_0$  and  $f_1$ . For  $x < 0$  both oscillate; for  $x > 0$ ,  $Ai(x)$  decays while  $Bi(x)$  grows, so boundary conditions usually decide which combination is physically allowed.

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# Classifying the point $x_0$

Return to  $y'' + P(x)y' + Q(x)y = 0$  and  $z = x - x_0$ . The earlier test asks whether  $P, Q$  are analytic (ordinary) or whether multiplying by  $z, z^2$  removes their singularities (regular singular). At an ordinary or regular singular point, the following limits exist. Put

$$p_0 = \lim_{z \rightarrow 0} zP(x), \quad q_0 = \lim_{z \rightarrow 0} z^2 Q(x).$$

Multiplying the equation by  $z^2$ , its leading terms are

$$z^2 y'' + p_0 z y' + q_0 y \approx 0.$$

Substitute  $y \approx a_0 z^r$ . Since  $y' \approx r a_0 z^{r-1}$  and  $y'' \approx r(r-1) a_0 z^{r-2}$ , the coefficient of  $a_0 z^r$  gives

$$r(r-1) + p_0 r + q_0 = 0. \quad (3.2.1)$$

Higher powers determine the recurrence, as in the examples already worked. The useful conclusions are:

- At an ordinary point there are two Taylor solutions.
- At a regular singular point there is at least one Frobenius solution. When two real roots differ by an integer, the larger root is the guaranteed starting point; the recurrence decides whether the smaller root also gives a pure Frobenius series.



# Classifying the point $x_0$ (cont.)

- A noninteger root gap gives two Frobenius solutions. Repeated roots require a logarithmic second solution. For a nonzero integer gap, inspect the recurrence: a logarithm may or may not be needed.
- At an irregular singular point, there is no general Frobenius guarantee.

The Taylor series, or the power-series factor of a Frobenius solution, converges at least up to the nearest other singularity of the normalized equation. We work on  $z > 0$  so that  $z^r = e^{r \ln z}$  is unambiguous. On the other side of  $x_0$ , rederive the equation in the coordinate  $s = x_0 - x > 0$ ; do not simply replace  $z$  by  $|z|$  in a finished solution. For complex  $z$ , choose a consistent branch.

# Where the logarithm in the second solution comes from

A logarithm appears when finding the second solution requires integrating  $1/x$ . The simplest example is

$$x^2 y'' + xy' = 0, \quad x > 0.$$

Trying  $y = x^r$  gives  $r(r-1) + r = r^2 = 0$ , a repeated root. One solution is  $y_1 = 1$ . To find another, divide by  $x$  and recognize a derivative:

$$xy'' + y' = (xy')' = 0.$$

Integrating twice gives

$$xy' = C_1, \quad y' = \frac{C_1}{x}, \quad y = C_0 + C_1 \ln x.$$

So the independent second solution is  $\ln x$ , not another power of  $x$ .

More generally, a repeated indicial root requires a logarithmic second solution. For distinct roots with an integer gap, a logarithm may or may not occur. The Euler example below has no logarithm; integer-order Bessel functions do. Reduction of order will provide a direct way to find the missing solution.

# Examples: ordinary and irregular singular points

**Airy equation.** For

$$y'' - xy = 0,$$

we have

$$P(x) = 0, \quad Q(x) = -x.$$

Both are analytic at  $x = 0$ , so the origin is an ordinary point. The two Taylor solutions  $f_0$  and  $f_1$  found earlier are therefore expected.

**The equation**  $x^4 y'' + 2x^3 y' + y = 0$ . For  $x \neq 0$ , divide by  $x^4$ :

$$y'' + \frac{2}{x} y' + \frac{1}{x^4} y = 0.$$

Thus

$$P(x) = \frac{2}{x}, \quad Q(x) = \frac{1}{x^4}.$$

Although

$$xP(x) = 2$$

is analytic, the regular-singular test fails because

$$x^2 Q(x) = \frac{1}{x^2}$$

## Examples: ordinary and irregular singular points (cont.)

is not analytic at the origin. Hence  $x = 0$  is an irregular singular point.

The lowest-power balance explains why a Frobenius series fails. If its first term is  $a_0 x^r$  with  $a_0 \neq 0$ , then

$$x^4 y'' + 2x^3 y' = a_0 r(r+1)x^{r+2} + \text{higher powers}, \quad y = a_0 x^r + \text{higher powers}.$$

Only the last term supplies  $x^r$ , so its coefficient cannot cancel. It would require  $a_0 = 0$ , contrary to the assumed leading term.

A change of variables confirms the diagnosis. Set  $u = 1/x$  and write  $y(x) = Y(u)$ . Since

$$y' = -u^2 Y_u, \quad y'' = u^4 Y_{uu} + 2u^3 Y_u,$$

the differential equation becomes

$$Y_{uu} + Y = 0.$$

Thus

$$y(x) = c_1 \cos\left(\frac{1}{x}\right) + c_2 \sin\left(\frac{1}{x}\right).$$

These solutions oscillate infinitely often near the origin and cannot be represented there by a power times a Taylor series.

## Example: an Euler equation with an integer root gap

Consider  $x^2 y'' - 6y = 0$  for  $x > 0$ . In normalized form,

$$y'' - \frac{6}{x^2}y = 0,$$

so  $xP(x) = 0$  and  $x^2Q(x) = -6$ : the origin is regular singular. Here a pure power already solves the full equation. Substitute

$$y = x^r, \quad y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}.$$

Then

$$x^2 y'' - 6y = [r(r-1) - 6]x^r = 0,$$

so  $(r-3)(r+2) = 0$ . Therefore

$$y_1 = x^3, \quad y_2 = x^{-2}, \quad y = c_1 x^3 + c_2 x^{-2}.$$

The roots differ by the integer 5, yet both solutions are pure powers. An integer gap is a reason to check the second solution, not proof that a logarithm must occur.

## Example: Bessel's regular singular point

Divide Bessel's equation by  $x^2$ :

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0.$$

The test gives  $xP = 1$  and  $x^2Q = x^2 - m^2$ , both analytic at zero. Thus the point is regular singular, with

$$p_0 = 1, \quad q_0 = -m^2.$$

The general indicial formula (3.2.1) immediately gives

$$r(r-1) + p_0r + q_0 = r(r-1) + r - m^2 = r^2 - m^2 = 0.$$

This recovers  $r = \pm m$  from the earlier direct substitution. Classification predicts that Frobenius applies; the recurrence still decides what happens at an exceptional index in the smaller-root solution.

# Summary: when Frobenius works

For  $y'' + P(x)y' + Q(x)y = 0$  near  $x_0$ :

- If  $P$  and  $Q$  are analytic,  $x_0$  is an ordinary point and there are two independent Taylor-series solutions.
- If  $P$  or  $Q$  fails to be analytic at  $x_0$ , but  $(x - x_0)P(x)$  and  $(x - x_0)^2Q(x)$  are analytic, then  $x_0$  is a regular singular point and at least one Frobenius solution exists.
- A noninteger root gap gives two Frobenius series. Repeated roots require a logarithmic second solution. For distinct roots separated by an integer, the recurrence decides whether a logarithm is needed.
- At an irregular singular point, Frobenius has no general existence guarantee. The lowest-power coefficient may directly contradict  $a_0 \neq 0$ .

A zero denominator in a recurrence must be returned to the undivided coefficient equation and checked for consistency.

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# Reduction of order: building the second solution

**Step 1: eliminate the unknown  $Q$  term.** Let  $P$  and  $Q$  be continuous on an interval  $I$ , and suppose a known solution  $v$  of

$$u'' + P(x)u' + Q(x)u = 0$$

is nonzero on the subinterval under consideration. Seek another solution  $y$ . The two equations are

$$y'' + Py' + Qy = 0, \quad v'' + Pv' + Qv = 0.$$

Multiply the first by  $v$ , the second by  $y$ , and subtract:

$$\begin{aligned} 0 &= v(y'' + Py' + Qy) - y(v'' + Pv' + Qv) \\ &= vy'' - yv'' + P(vy' - yv') + Q(vy - yv). \end{aligned}$$

The  $Q$  term vanishes because  $vy - yv = 0$ . Hence

$$vy'' - yv'' + P(vy' - yv') = 0.$$

Define the Wronskian in the order

$$W[v, y] = vy' - v'y = vy' - yv'.$$

# Reduction of order: building the second solution (cont.)

Its derivative is

$$\begin{aligned}W'[v, y] &= (vy' - v'y')' \\&= v'y' + vy'' - v''y - v'y' \\&= vy'' - yv''.\end{aligned}$$

Therefore

$$W' + PW = 0. \quad (3.3.1)$$

**Step 2: solve the Wronskian equation.** Choose a reference point  $x_* \in I$  and define the integrating factor

$$\mu(x) = \exp\left(\int_{x_*}^x P(t) dt\right).$$

By the fundamental theorem of calculus and the chain rule,

$$\mu'(x) = P(x)\mu(x).$$

Multiplying (3.3.1) by  $\mu$  gives

$$\begin{aligned}0 &= \mu W' + \mu PW \\&= \mu W' + \mu' W \\&= (\mu W)'. \end{aligned}$$

## Reduction of order: building the second solution (cont.)

Thus

$$\mu(x)W(x) = C_1,$$

or

$$W[v, y](x) = C_1 \exp\left(-\int_{x_*}^x P(t) dt\right). \quad (3.3.2)$$

This is Abel's identity for the Wronskian.

**Step 3: integrate once more to recover  $y$ .** Next observe that

$$\begin{aligned}\left(\frac{y}{v}\right)' &= \frac{vy' - yv'}{v^2} \\ &= \frac{W[v, y]}{v^2}.\end{aligned}$$

Using (3.3.2),

$$\left(\frac{y}{v}\right)' = C_1 \frac{\exp\left(-\int_{x_*}^x P(t) dt\right)}{v^2(x)}.$$

## Reduction of order: building the second solution (cont.)

Integrate from  $x_*$  to  $x$ :

$$\frac{y(x)}{v(x)} = C_0 + C_1 \int_{x_*}^x \frac{\exp\left(-\int_{x_*}^s P(t) dt\right)}{v^2(s)} ds.$$

The constant  $C_0$  contributes only a multiple of  $v$ . Choose  $C_0 = 0$ ,  $C_1 = 1$  to obtain a second solution:

$$y_2(x) = v(x) \int_{x_*}^x \frac{\exp\left(-\int_{x_*}^s P(t) dt\right)}{v^2(s)} ds. \quad (3.3.3)$$

Writing  $y_2 = vI$ , the product rule gives

$$W[v, y_2] = v(v'I + vI') - v'(vI) = v^2 I' = \exp\left(-\int_{x_*}^x P(t) dt\right) \neq 0.$$

Thus the constructed solution is independent of  $v$ .

# Reduction-of-order example: harmonic oscillator

For  $y'' + y = 0$ , take the known solution  $v(x) = \cos x$ . Here  $P(x) = 0$ . Work on  $(-\pi/2, \pi/2)$ , where  $v \neq 0$ , and choose  $x_* = 0$ . The exponential in (3.3.3) is  $\exp(-\int_0^x 0 dt) = 1$ , so

$$\begin{aligned}y_2(x) &= \cos x \int_0^x \frac{ds}{\cos^2 s} = \cos x \int_0^x \sec^2 s ds \\&= \cos x [\tan s]_0^x = \cos x \tan x = \sin x.\end{aligned}$$

The independence check uses the derivatives of sine and cosine:

$$W[\cos x, \sin x] = \cos x \cos x - (-\sin x) \sin x = \cos^2 x + \sin^2 x = 1.$$

Thus reduction of order recovers the familiar second solution.

# Reduction-of-order example: Bessel's second solution

For Bessel's equation in normalized form,

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0,$$

we have

$$P(x) = \frac{1}{x}, \quad v(x) = J_m(x).$$

Choose an interval contained in  $(0, \infty)$  on which  $J_m$  has no zero. For a reference point  $x_* > 0$ ,

$$\begin{aligned} \exp\left(-\int_{x_*}^s \frac{dt}{t}\right) &= \exp\left[-\log\left(\frac{s}{x_*}\right)\right] \\ &= \frac{x_*}{s}. \end{aligned}$$

The constant factor  $x_*$  can be absorbed into the arbitrary normalization. Thus a second solution is

$$\tilde{Y}_m(x) = J_m(x) \int_{x_*}^x \frac{ds}{sJ_m^2(s)}. \quad (3.3.4)$$

## Reduction-of-order example: Bessel's second solution (cont.)

Its Wronskian follows directly from the definition. Writing  $\tilde{Y}_m = J_m I$  with  $I(x) = \int_{x_*}^x \frac{ds}{s J_m^2(s)}$ , so that  $I'(x) = \frac{1}{x J_m^2(x)}$ ,

$$W[J_m, \tilde{Y}_m] = J_m(J'_m I + J_m I') - J'_m J_m I = J_m^2 I' = \frac{1}{x}.$$

The standard function  $Y_m$  is normalized so that  $W[J_m, Y_m] = 2/(\pi x)$ . Its full definition is given in Chapter 5. Consequently

$$Y_m(x) = \frac{2}{\pi} \tilde{Y}_m(x) + C J_m(x). \quad (3.3.5)$$

The Wronskian fixes the scale but not the additive multiple of  $J_m$ . Changing the lower integration limit changes only  $C$ .

The singular behavior at the origin follows directly from the integral. For  $m \geq 1$ ,

$$J_m(x) \sim \frac{x^m}{2^m m!} \quad (x \rightarrow 0^+).$$

# Reduction-of-order example: Bessel's second solution (cont.)

Hence

$$\begin{aligned}\frac{1}{sJ_m^2(s)} &\sim \frac{1}{s} \left( \frac{2^m m!}{s^m} \right)^2 \\ &= 2^{2m} (m!)^2 s^{-2m-1}.\end{aligned}$$

Integrating toward the origin,

$$\int_{x_*}^x \frac{ds}{sJ_m^2(s)} \sim 2^{2m} (m!)^2 \int_{x_*}^x s^{-2m-1} ds = 2^{2m} (m!)^2 \left[ -\frac{s^{-2m}}{2m} \right]_{x_*}^x.$$

The lower-limit contribution stays bounded. Keeping the singular term and multiplying by  $J_m(x) \sim x^m/(2^m m!)$  gives

$$\tilde{Y}_m(x) \sim \frac{x^m}{2^m m!} \left[ -\frac{2^{2m} (m!)^2}{2m} x^{-2m} \right] = -2^{m-1} (m-1)! x^{-m}.$$

After the factor  $2/\pi$  in (3.3.5),

$$Y_m(x) \sim -\frac{(m-1)!}{\pi} \left( \frac{2}{x} \right)^m, \quad m \geq 1.$$



# Reduction-of-order example: Bessel's second solution (cont.)

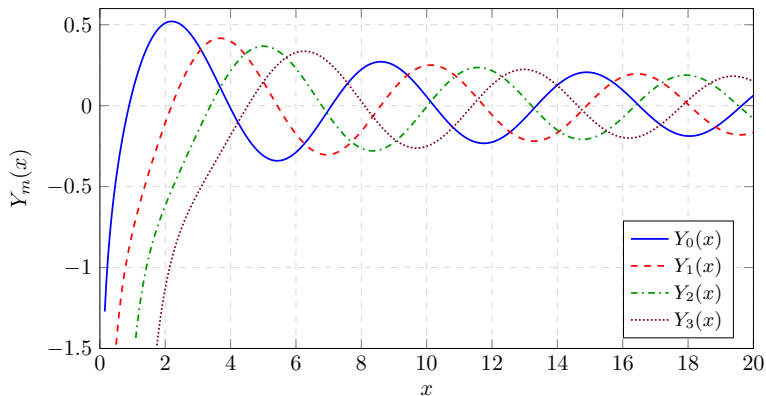
For  $m = 0$ ,  $J_0(s) = 1 + O(s^2)$ , so the integral instead begins with  $1/s$ :

$$\frac{1}{sJ_0^2(s)} = \frac{1}{s} + O(s), \quad \int_{x_*}^x \frac{ds}{sJ_0^2(s)} = \log x + C_* + O(x^2).$$

Multiplication by  $J_0(x) \rightarrow 1$  gives  $\tilde{Y}_0(x) \sim \log x$  and hence  $Y_0(x) \sim (2/\pi) \log x$ . Chapter 5 supplies the conventional additive constant. Here the key point is the mechanism: the missing solution comes from integrating  $1/x$ .

Thus every  $Y_m$  is singular at the origin.

# The Bessel functions of the second kind $Y_0, \dots, Y_3$



Unlike  $J_m$ , each  $Y_m$  plunges to  $-\infty$  as  $x \rightarrow 0^+$ : the second Bessel solution is singular at the center, so it is excluded for a drum (finite there) but needed on annular domains.

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# The Airy Wronskian and a useful primitive

The power series built  $Ai$  and  $Bi$  near the origin. Applications—such as the radial field near a curved reflecting surface in later wave courses—need global facts: which solutions decay, where the zeros sit, and which combination is “outgoing.” Three tools from earlier chapters supply them.

**Step 1: the Wronskian is constant.** For  $y'' - xy = 0$  the normalized coefficient is  $P = 0$ , so Abel's identity (3.3.2) makes the Wronskian of any solution pair constant. Evaluating  $W[Ai, Bi] = Ai Bi' - Ai' Bi$  at  $x = 0$  with the initial values (3.1.15) and  $Bi(0) = \sqrt{3} Ai(0)$ ,  $Bi'(0) = -\sqrt{3} Ai'(0)$  gives

$$W[Ai, Bi] = -2\sqrt{3} Ai(0)Ai'(0) = \frac{2\sqrt{3}}{3\Gamma(2/3)\Gamma(1/3)}.$$

Euler's reflection formula (Chapter 2) gives  $\Gamma(1/3)\Gamma(2/3) = \pi/\sin(\pi/3) = 2\pi/\sqrt{3}$ , so

$$W[Ai, Bi](x) = \frac{1}{\pi}. \quad (3.4.1)$$

**Step 2: a primitive for the squared solution.** If  $w'' = tw$  for a complex argument  $t$ , then differentiating gives

$$\frac{d}{dt} [t w(t)^2 - w'(t)^2] = w^2 + 2tw w' - 2w'(tw) = w(t)^2.$$

# The Airy Wronskian and a useful primitive (cont.)

Hence

$$\int w(t)^2 dt = t w(t)^2 - w'(t)^2 + C, \quad w'' = tw, \quad (3.4.2)$$

an exact antiderivative for the square of *any* Airy solution. When a mode norm  $\int w^2$  must be evaluated—in residue or perturbation formulas, for example—(3.4.2) reduces it to endpoint data involving only  $w$  and  $w'$ .

# Airy's Fourier derivation and integral representations

A Fourier transform obtains the Airy integral directly from the differential equation. Use the Fourier convention

$$\widehat{y}(t) = \int_{-\infty}^{\infty} y(x) e^{-itx} dx, \quad y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{y}(t) e^{itx} dt.$$

For functions with vanishing boundary terms, two integrations by parts give

$$\mathcal{F}\{y''\}(t) = -t^2 \widehat{y}(t).$$

Differentiating the transform with respect to  $t$  gives

$$\widehat{y}'(t) = \int_{-\infty}^{\infty} (-ix) y(x) e^{-itx} dx = -i \mathcal{F}\{xy\}(t),$$

and therefore

$$\mathcal{F}\{xy\}(t) = i \widehat{y}'(t).$$

The sign is positive; it controls the phase of the transformed solution below. Transforming  $y'' - xy = 0$  now gives

$$-t^2 \widehat{y} - i \widehat{y}' = 0, \quad \text{so} \quad \frac{\widehat{y}'}{\widehat{y}} = it^2.$$

# Airy's Fourier derivation and integral representations (cont.)

Integration yields

$$\widehat{y}(t) = Ce^{it^3/3}.$$

Apply the inverse transform. On a symmetric interval, the negative- $t$  integrand is the complex conjugate of the positive- $t$  integrand, so

$$\begin{aligned} y(x) &= \frac{C}{2\pi} \int_{-\infty}^{\infty} e^{i(t^3/3+xt)} dt \\ &= \frac{C}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + xt\right) dt. \end{aligned}$$

Choosing  $C = 1$  gives

$$Ai(x) = \frac{1}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + xt\right) dt. \quad (3.4.3)$$

This is the same  $Ai$  normalized in (3.1.15). Indeed, set  $u = t^3/3$ . The differential and the factor needed for the derivative at  $x = 0$  become

$$dt = 3^{-2/3} u^{-2/3} du, \quad t dt = 3^{-1/3} u^{-1/3} du.$$

# Airy's Fourier derivation and integral representations (cont.)

The sector-rotation calculation from Chapter 2 gives, for  $0 < \alpha < 1$ ,

$$\int_0^\infty u^{\alpha-1} \cos u \, du = \Gamma(\alpha) \cos\left(\frac{\pi\alpha}{2}\right), \quad \int_0^\infty u^{\alpha-1} \sin u \, du = \Gamma(\alpha) \sin\left(\frac{\pi\alpha}{2}\right).$$

Use  $\alpha = 1/3$  in the cosine integral and  $\alpha = 2/3$  in the sine integral. Euler's reflection formula gives  $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$ , so

$$\begin{aligned} \frac{1}{\pi} \int_0^\infty \cos(t^3/3) \, dt &= \frac{3^{-2/3}}{\pi} \Gamma(1/3) \cos(\pi/6) = \frac{1}{3^{2/3} \Gamma(2/3)}, \\ -\frac{1}{\pi} \int_0^\infty t \sin(t^3/3) \, dt &= -\frac{3^{-1/3}}{\pi} \Gamma(2/3) \sin(\pi/3) = -\frac{1}{3^{1/3} \Gamma(1/3)}. \end{aligned}$$

These are precisely  $Ai(0)$  and  $Ai'(0)$  in (3.1.15); uniqueness for the initial-value problem identifies the integral solution with the standard  $Ai$ . The integral is oscillatory, not absolutely convergent. It is understood as the limit of truncated integrals, or equivalently through an Abel convergence factor. This is the Fourier route from the differential equation to  $Ai$ .



# Airy's Fourier derivation and integral representations (cont.)

The real Fourier contour selects the solution that decays for  $x > 0$ . The independent companion  $Bi$ , which grows there, has the real-axis representation

$$Bi(x) = \frac{1}{\pi} \int_0^{\infty} \left[ e^{-t^3/3+xt} + \sin \left( \frac{t^3}{3} + xt \right) \right] dt.$$

The first term supplies the exponentially growing branch for  $x > 0$ ; for  $x < 0$  it is smaller than the oscillatory term at large  $|x|$ . The sine integral is understood in the same oscillatory sense as (3.4.3).

To justify the Fourier integrations by parts for  $Ai$ , introduce

$$F_{\varepsilon}(X) = \int_{-\infty}^{\infty} Ai(t) e^{(\varepsilon - iX)t} dt, \quad \varepsilon > 0.$$

# Airy's Fourier derivation and integral representations (cont.)

The factor damps the negative tail, while Airy decay controls the positive tail, so the boundary terms vanish. From  $Ai''(t) = tAi(t)$ ,

$$\begin{aligned}(\varepsilon - iX)^2 F_\varepsilon(X) &= \int_{-\infty}^{\infty} Ai''(t) e^{(\varepsilon - iX)t} dt \\ &= \int_{-\infty}^{\infty} t Ai(t) e^{(\varepsilon - iX)t} dt = iF'_\varepsilon(X).\end{aligned}$$

Solving this first-order equation gives

$$F_\varepsilon(X) = F_\varepsilon(0) \exp\left(\frac{(\varepsilon - iX)^3 - \varepsilon^3}{3}\right).$$

To fix the remaining transform constant, use the standard Airy normalization  $\int_{-\infty}^{\infty} Ai(t) dt = 1$  (quoted). Thus  $F_\varepsilon(0) \rightarrow 1$  as  $\varepsilon \rightarrow 0^+$ , and removing the Abel factor yields

$$\int_{-\infty}^{\infty} Ai(t) e^{-iXt} dt = e^{iX^3/3}, \quad \int_{-\infty}^{\infty} Ai(t) dt = 1. \quad (3.4.4)$$

# Airy's Fourier derivation and integral representations (cont.)

The second equality is an ordinary conditionally convergent improper integral; the Abel limit has the same value.

# Large-argument behavior by stationary phase

**Negative argument: oscillation.** For  $x = -q$  with  $q \rightarrow +\infty$ , write (3.4.3) as the real part of  $J(q) = \int_0^\infty e^{i(t^3/3 - qt)} dt$ . The phase  $\phi(t) = t^3/3 - qt$  has  $\phi'(t) = t^2 - q$ : one interior stationary point  $t_0 = \sqrt{q}$ , with

$$\phi(t_0) = -\frac{2}{3}q^{3/2}, \quad \phi''(t_0) = 2\sqrt{q} > 0.$$

The stationary-phase rule of Chapter 1 (the endpoint  $t = 0$  contributes only  $O(1/q)$ ) gives

$$J(q) \sim e^{-2iq^{3/2}/3} e^{i\pi/4} \sqrt{\frac{2\pi}{2\sqrt{q}}} = \sqrt{\pi} q^{-1/4} e^{i(\pi/4 - 2q^{3/2}/3)}.$$

Taking the real part for  $Ai$  and the imaginary part for  $Bi$ —its companion integral  $\frac{1}{\pi} \int_0^\infty e^{-t^3/3 - qt} dt$  is endpoint-dominated and smaller than  $q^{-1/4}$ —gives, with  $\chi(q) = \frac{2}{3}q^{3/2} + \frac{\pi}{4}$ ,

$$Ai(-q) \sim \frac{q^{-1/4}}{\sqrt{\pi}} \sin \chi, \quad Bi(-q) \sim \frac{q^{-1/4}}{\sqrt{\pi}} \cos \chi, \quad q \rightarrow +\infty. \quad (3.4.5)$$

Both oscillate with a slowly decaying  $q^{-1/4}$  envelope and a phase advancing like  $q^{3/2}$ .

# Large-argument behavior by stationary phase (cont.)

**Positive argument: exponential split.** Now  $\phi'(t) = t^2 + x$  has no real zero: the stationary points move to  $t = \pm i\sqrt{x}$ , where  $e^{i\phi} = e^{\mp 2x^{3/2}/3}$ . The decaying and growing branches pick out different saddles:

$$Ai(x) \sim \frac{x^{-1/4}}{2\sqrt{\pi}} e^{-2x^{3/2}/3}, \quad Bi(x) \sim \frac{x^{-1/4}}{\sqrt{\pi}} e^{+2x^{3/2}/3}, \quad x \rightarrow +\infty. \quad (3.4.6)$$

We quote these; they follow by deforming (3.4.3) onto the complex saddle, the full steepest-descent case of Chapter 1.

**Why a special function was needed.** As  $x \rightarrow 0$  the two saddles  $t = \pm\sqrt{-x}$  coalesce: no single Gaussian or exponential survives, and the Airy function itself is the uniform object bridging the two regimes. Wave problems call such a place a turning point or a shadow/caustic transition; (3.4.5)–(3.4.6) are its two asymptotic faces.

# Zeros, rotated arguments, and the outgoing pair

**Zeros.** All zeros of  $Ai$  and of  $Ai'$  lie on the negative real axis. Writing  $Ai(-q_p) = 0$  and  $Ai'(-q'_p) = 0$ , the first few are

$$q_p : 2.3381, 4.0879, 5.5206, \dots \quad q'_p : 1.0188, 3.2482, 4.8201, \dots$$

The asymptotic (3.4.5) estimates them without numerics:  $\sin \chi = 0$  needs  $\chi \approx p\pi$ , giving  $q_p \approx [3\pi(4p-1)/8]^{2/3}$ —at  $p = 1$  this is 2.32 against the exact 2.3381. The zeros grow like  $p^{2/3}$ .

**Rotating the argument.** If  $y''(x) = xy(x)$  and  $w(z) = y(ze^{i\theta})$ , the chain rule gives  $w''(z) = e^{3i\theta} z w(z)$ : the rotated function is again an Airy solution exactly when  $e^{3i\theta} = 1$ , i.e.  $\theta = \pm 2\pi/3$ . Matching  $Ai(ze^{-2\pi i/3}) = \alpha Ai(z) + \beta Bi(z)$  at  $z = 0$  through (3.1.15) gives  $\alpha + \sqrt{3}\beta = 1$  and  $\alpha - \sqrt{3}\beta = e^{-2\pi i/3}$ , hence  $\alpha = \frac{1}{2}e^{-i\pi/3}$ ,  $\beta = \frac{1}{2}e^{i\pi/6}$ :

$$Ai(ze^{-2\pi i/3}) = \frac{1}{2} e^{-i\pi/3} [Ai(z) + iBi(z)]. \quad (3.4.7)$$

The conjugate identity has  $e^{+i\pi/3}$  and  $Ai - iBi$ .

**The outgoing combination.** Rearranging (3.4.7),

$$Bi(z) - iAi(z) = 2e^{-i\pi/6} Ai(ze^{-2\pi i/3}). \quad (3.4.8)$$

## Zeros, rotated arguments, and the outgoing pair (cont.)

The diffraction literature names the two conjugate combinations

$$w_1(z) = \sqrt{\pi} [Bi(z) + iAi(z)], \quad w_2(z) = \sqrt{\pi} [Bi(z) - iAi(z)] \quad (3.4.9)$$

the Fock–Airy functions: they play for Airy's equation the role that  $H^{(1)} = J + iY$  and  $H^{(2)} = J - iY$  play for Bessel's. Their Wronskian follows from (3.4.1):

$$w_1 w_2' - w_1' w_2 = 2\pi i W[Ai, Bi] = 2i.$$

Since (3.4.8) reads  $w_2(z) = 2\sqrt{\pi} e^{-i\pi/6} Ai(ze^{-2\pi i/3})$ , the zeros of  $w_2$  sit on the ray  $\arg z = -\pi/3$ , at  $z_p = q_p e^{-i\pi/3}$ . In the wave applications these rotated zeros are precisely the complex poles—the creeping-wave poles—of the outgoing radial problem.

# Chapter summary

- A Frobenius expansion has the form  $y = \sum_{k=0}^{\infty} a_k(x - x_0)^{k+r}$  with  $a_0 \neq 0$ .
- The lowest-power coefficient gives the indicial equation. Higher coefficients give a recurrence, but every zero recurrence denominator must be checked in the original, undivided coefficient equation.
- The harmonic oscillator produces the even and odd Taylor chains  $\cos x$  and  $\sin x$ .
- Bessel's equation has indicial roots  $\pm m$ . The regular root gives  $J_m$ ; for positive integer  $m$ , the negative-root branch fails a compatibility equation, and the missing solution is  $Y_m$ .
- Airy's equation (ordinary point) has two Taylor solutions  $f_0, f_1$  generated by  $y(0)$  and  $y'(0)$ . The classical  $A_i$  and  $B_i$  are the linear combinations with the standard initial values (3.1.15). Their global tools are  $W[A_i, B_i] = 1/\pi$  (3.4.1), the primitive (3.4.2), the oscillatory/exponential asymptotics (3.4.5)–(3.4.6), and the outgoing pair  $w_{1,2}$  (3.4.9), with the zeros of  $w_2$  on  $\arg z = -\pi/3$ .
- Ordinary points have two analytic solutions. Regular singular points have at least one Frobenius solution; an integer root gap may introduce a logarithm or a compatibility degeneracy. Irregular singular points have no general Frobenius guarantee.
- Reduction of order gives

$$y_2 = v \int \frac{e^{-\int P}}{v^2} dx,$$

with consistent limits understood; its nonzero Wronskian proves independence from  $v$ .